Optical Polarization: Malus' Law, Sugar Concentration Determination and Faraday Effect

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In this paper, we examine the usage of polarizers, $\lambda/4$ and $\lambda/2$ wave plates, as well as Brewster reflections and calcite to produce and detect polarized light. Malus' law, which states a cosine squared dependence of the intensity, is verified. The birefringence of calcite is demonstrated, and the orthogonal ordinary and extraordinary beams are identified. The Jones formalism is introduced to provide a systematic description of polarized light and the mentioned optical elements. Using a semi-shadow polarimeter, the rotation angle of sugar was determined as a function of wavelength to directly measure the sugar concentration in water. Furthermore, the Faraday effect is investigated using a $(3,044 \pm 0,005)$ cm glass cylinder to calculate the Verdet constants for distinct wavelength ranges. As expected, these constants decrease with increasing wavelength, with one exception.

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1 Introduction

Polarization is an essential property of transverse waves such as light and plays a central role in many modern applications, ranging from photography and optical microscopy to telecommunication, sunglasses, and the analysis of molecular structures. This experiment focuses on how polarized light is generated and can be modified using various optical components. Simple polarizers, birefringent crystals, such as calcite, and wave plates are used to produce and control specific polarization states. The chiral behavior of sugar solutions is investigated with a half-shade polarimeter to quantify the rotation of the plane of polarization. Furthermore, the Faraday effect is examined as a second mechanism for rotating polarized light, where an external magnetic field causes a magneto-optical rotation of the polarization direction.

2 Malus's law

2.1 Theory of Malus's law

Malus's law describes the ratio of the intensity I of a linearly polarized electromagnetic wave after passing through an ideal polarizing filter to the original intensity I_0. The transmission is a function of the angle θ, which describes the orientation of the polarizer's optical axis relative to the wave's original polarization direction. Malus's law is described by Equation [1].

$$I(\theta) = I_0 * cos^2(\theta) \tag{1}$$

After passing through the polarizing filter, the remaining wave is polarized along the filter axis. The energy components that are not transmitted are absorbed by the polarizing filter and reflected when a polarizing beam splitter is used.

2.2 Experimental setup

The unpolarized light of an incandescent lamp was vertically polarized with a polarizer. The polarized light then falls onto an analyzer. The transmission axis of the analyzer and that of the polarizer enclose the angle θ. The intensity of the transmitted light was measured using a photocell. According to the theory, the intensity of light should behave according to Malus's law [1].

2.3 Evaluation of the measured data

I_0 is the intensity of the polarized light falling in the analyzer. To check whether the intensity changes according to Malus's law, the angle θ was varied in increments of $\Delta\theta = 10°$, and the intensity $I(\theta)$ was recorded after the analyzer. The results were presented in figure [1]. The graph shows that the measured values match the theoretical values. The angles of 0°-180° were set in the analyzer. Figure [1] shows that the scaling of the analyzer is shifted by 15°. The measured values were taken from $\theta_{min} = 15$

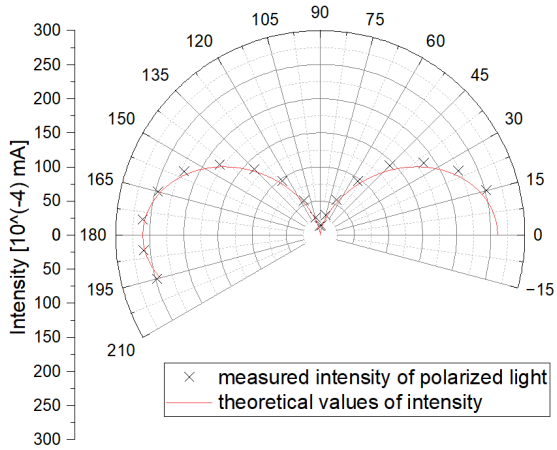

Figure 1: Measured intensities of polarized light as a function of the angle θ between the polarizer and analyzer.

to $\theta_{max} = 195$. The step size is $\Delta\theta = 10$. The correction has already been taken in the plot. The graph shows that the values never reach 0, even though the transmission axes are perpendicular to each other. This is due to the remaining stray light that was detected in the experimental setup by the photocell from residual light sources.

3 Double refraction in the calcite crystal

3.1 Motivation

The aim of this experiment is to investigate the birefringence of unpolarized light as it passes through an anisotropic calcite crystal. It will be shown that the light splits into an ordinary and an extraordinary beam. These beams have different propagation properties. The experiment will demonstrate that both partial beams are linearly polarized but perpendicular to each other.

3.2 Evaluation

The unpolarized light beam from an incandescent lamp falls on a calcite crystal, is split into ordinary and extraordinary beams, and is projected onto a screen. The experiment was set up so that no light from the incandescent lamp falls directly on the screen but only the light passing through the calcite crystal is projected. A polarization filter is used as an analyzer to identify the polarization of the two beams. In this process, one beam showed a minimum intensity at $\theta_1 = (50,0 \pm 1,0)^\circ$ and the other beam a minimum at $\theta_2 = (140,0 \pm 1,0)^\circ$. The difference $\Delta\theta = (90 \pm 1,4)^\circ$ shows that the light rays are polarized perpendicular to each other. This result is also expected according to theory. To identify the ordinary or extraordinary ray, one can rotate the calcite crystal. The ordinary ray should stay in one place, while the extraordinary ray also rotates.

4 λ-quarter plate and λ-half plate

A lambda-quarter plate consists of birefringent crystals in which the propagation speed differs along the fast and slow axes. As a result, two electromagnetic waves polarized perpendicular to each other experience a different phase shift. If linearly polarized light is shone onto a quarter-wave plate, the polarization of the emerging light depends on the angle between the orientation of the incoming light and the axis of the plate.
For $\alpha = [0, \frac{\pi}{2}]$ the light remains linearly polarized because the light is polarized parallel to one of the optical axes and excites only that axis. Thus, no relative phase shift occurs. For $\alpha = \frac{\pi}{4}$ both axes are excited equally. A phase shift of $\Delta\Phi = \frac{\pi}{2}$ arises, so the emerging light is circularly polarized. At any angle between $\alpha = [0, \frac{\pi}{2}]$, elliptically polarized light is produced, since the ratio is not balanced. This can be understood using Malus' law [1]. To determine the polarization states, a polarizer was used as an analyzer. Linearly polarized light is completely extinguished at a certain angular setting of the analyzer. This occurs when the transmission axis is perpendicular to the polarization. For elliptically polarized light, one can observe an increase and decrease in intensity, but no complete extinction. In the special case of circularly polarized light, the intensity remains constant regardless of the angular setting.
If two identical quarter-wave plates are connected in series with their optical axes aligned parallel, the phase shifts can be added. The system is equivalent to a half-wave plate. The phase shift experienced by an electromagnetic wave passing through this system is $\Delta\Phi = \pi$. A half-wave plate rotates the direction of linear polarization, thereby changing the polarization state of linearly polarized light. The rotation angle is twice the angle between the incident polarization and the optical axis of the plate. The experimental setup demonstrates exactly this behavior. The light remains linearly polarized, but with a rotated polarization plane.
If the quarter-wave plate made of quartz is replaced with a plate made of calcite, the thickness of the new plate must be adjusted to achieve the same result. The thickness of the plate is crucial because it determines the phase difference between the fast and slow axes. If the new plate meets all the requirements of a quarter-wave plate, the observable effects do not change.

5 Jones-Formalism

The Jones formalism offers the possibility to describe the polarization of light and the effect of polarizers and wave plates. The polarization states can be represented as a complex two-dimensional vector, describing the amplitudes of the x and y directions and the relative phase. The polarization states of x, y, circular right-handed, circular left-handed, and diagonal polarized are, therefore, given by

$$|\rightarrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \tag{2}$$

$$|\uparrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \tag{3}$$

$$|\nearrow\rangle = \frac{1}{\sqrt{2}} * (|\rightarrow\rangle + |\uparrow\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \tag{4}$$

$$|\sigma_+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} \tag{5}$$

$$|\sigma_-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix} \tag{6}$$

The polarizers in x, y and diagonal direction, as well as the quarter-wave plate are

$$\hat{P}_x = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \tag{7}$$

$$\hat{P}_y = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \tag{8}$$

$$\hat{P}_{diag} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \tag{9}$$

$$\hat{R}_{\frac{\lambda}{4}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1-i & 0 \\ 0 & 1+i \end{pmatrix} \tag{10}$$

These matrices can be systematically derived by considering the transformation of the basic polarization states by the respective optical element. Multiplying $\hat{R}_{\frac{\lambda}{4}}$ by itself, the effect of 4 can be understood.

$$\hat{R}^2_{\frac{\lambda}{4}} = \frac{1}{2} \begin{pmatrix} -2i & 0 \\ 0 & 2i \end{pmatrix} = -i\hat{R}_{\frac{\lambda}{2}} \tag{11}$$

The remaining factor $-i$ is negligible, since a global phase cannot be measured.

6 Determination of sugar concentration in water

6.1 Motivation

Accurate concentration analysis of unknown solutions is fundamental to advancements in research, industrial processing, and clinical diagnostics. To quickly estimate the concentration of a solution, polarized light of a specific wavelength, the penumbral angle at a known concentration, the penumbral angle of the solution under investigation, and equation [12] are used. The analysis was performed with a penumbral polarimeter.

6.2 Determination of the rotation angle $[\alpha]_\lambda$ of sugar

To determine the sugar concentration in water, a semi-shadow polarimeter is used. First, the semi-shadow angle of the semi-shadow polarimeter must be determined. This requires finding the angle at which the darkest state is reached between area I and area II. The penumbra division of the penumbra polarimeter can be seen in figure [2].

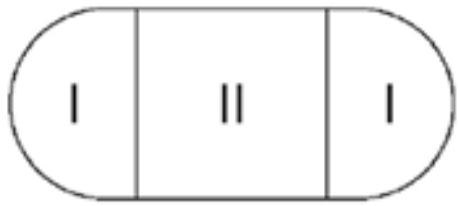

Figure 2: The penumbra division of the used penumbra polarimeter. The object is divided into areas I and II. In this experiment, the angle was alternately sought at which area I or II appears darkest.

For this purpose, five measurements were taken per area and the mean value with standard error was calculated. The semi-shadow angle of the semi-shadow polarimeter used is $\Psi = (10,36 \pm 0,61)$. The specific rotation of sugar must be determined before the sugar concentration in a solution can be determined. To determine the specific rotation at a fixed wavelength, a polarization filter is used. Then, the half-shadow angle for a specific wavelength is determined. The light passes through a sugar-water solution with a known concentration of $\rho_0 = (10,3 \pm 2,1) * 10^{-2} \frac{kg}{L}$. The glass cylinder containing this solution has a length of $l = (26,5 \pm 0,1)cm$. The measured half-shadow angles depending on different wavelengths can be found in table [1].

wavelength [nm]	Penumbra angle α [degree]
505	9,4±1,5
525	6,38±0,57
580	9,8±0,97

Table 1: Measured penumbral angles as a function of wavelength. The penumbral angles were recorded at a known sugar concentration ρ_0.

Using formula [12] and the known quantities, one

can calculate the specific rotation $[\alpha]_\lambda$.

$$[\alpha]_\lambda = \frac{\alpha(\lambda)}{\rho * l} \tag{12}$$

Figure [3] shows the specific rotation calculated from the values in Table [1] as a function of wavelength.

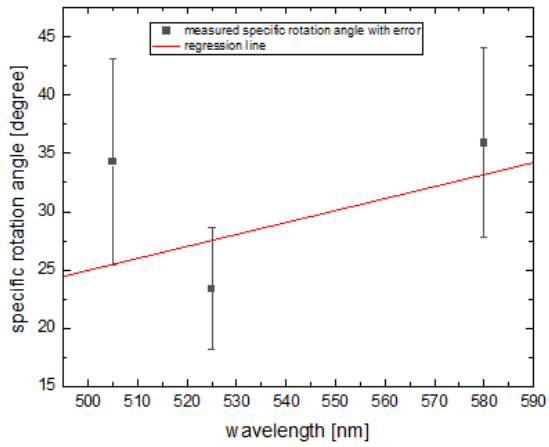

Figure 3: The calculated specific rotation angle of sugar as a function of wavelength. The assumption of a linear dispersion relationship could not be confirmed by linear regression.

Graphic [3] provides no information on a dispersion relation due to the large overlaps of the error bars.

6.3 Determination of sugar concentration in water

Using the calculated values from Graphic [3] and formula [12], one can now determine the concentration of sugar in water.
The solution with an unknown sugar concentration was illuminated with wavelength light $\lambda = 580nm$. A half-shadow polarimeter was used to measure a half-shadow angle of $\alpha = (8,24 \pm 0,40)$. Using the determined α and equation [12], a concentration of $\rho_1 = (8,7 \pm 2,0) * 10^{-2}\frac{kg}{L}$ was determined. Since the original solution with ρ_0 was diluted by adding $\Delta V_{water} = (19,4 \pm 3,0)ml$ of watB ist er, the concentration can be estimated as $\rho_{1-Lit} = (8,6 \pm 1,1) * 10^{-2}\frac{kg}{L}$. The values are consistent with each other.

7 Fraday Effect

The Faraday effect is a magneto-optical phenomenon that can be observed when light propagates through a medium, such as a glass cylinder, subjected to a strong external magnetic field.

$$\theta \propto B$$

The Formula for this effect is the following

$$\theta = VBl \tag{13}$$

B stands for the magnetic flux, l is the length of the glass cylinder, V is the Verdet constant and θ the angle of the polarization. The Verdet constant V characterizes how strongly a material rotates the polarization of light when its inside a strong magnetic field. It depends on the refractive index n of the material and on the wavelength λ of the light. The physical origin of the Faraday effect is a magnetically induced circular birefringence inside the medium. When an external magnetic field B is applied, the resonance frequencies of the bound electrons shift differently for left- and right-circularly polarized light. This frequency shift can be related to the normal Zeeman effect, which is sufficient to describe the phenomenon in transparent media.
In a simplified model, an electron bound in an atom behaves like a harmonic oscillator. When a magnetic field is present, its motion can be decomposed into a component parallel to B and a perpendicular oscillation. The perpendicular component corresponds to a combination of circularly polarized waves. Due to the Lorentz force, the orbital motion of the electron is modified, which leads to different effective oscillation frequencies for σ^+ and σ^- polarizations. Consequently, the refractive index differs slightly for the two helicities, which introduces a net rotation of the polarization plane.
The energy difference between the two circular components in the normal Zeeman effect is

$$\Delta E = \frac{\hbar e B}{m_e}, \tag{5}$$

where e is the elementary charge of the electron, m_e the electron mass, and $\hbar$ the reduced Planck constant.
Since the rotation angle θ is linked to the difference in refractive index between the two circular components, one may write

$$\theta = \frac{\pi\,\Delta n\,l}{\lambda}, \tag{6}$$

with $\Delta n = n_{\sigma^+} - n_{\sigma^-}$ and l the length of the medium. Because the refractive index varies with wavelength, the Zeeman-induced index difference can be approximated by the dispersion relation:

$$\Delta n_\lambda \approx \frac{dn}{d\lambda}.$$

Combining dispersion, Zeeman splitting, and the classical electron model yields

$$\theta = -\frac{e}{2m_e c\,\lambda}\,B\,l\,\frac{dn}{d\lambda}, \tag{7}$$

which bridges the microscopic interaction and the macroscopic optical rotation. Equation (7), when

compared to the phenomenological Faraday law $\theta = VBl$, allows one to extract the Verdet constant of the material.

$$V = -\frac{e}{2m_e c}\lambda\frac{dn}{d\lambda} \tag{8}$$

This is the resulting formula which depends on the dispersion relation of the glass $\frac{dn}{d\lambda}$ at the wavelength of the incoming light.

7.1 Determination of the Verdet Constant

7.1.1 Magnetic Field Calibration and Setup

The experimental procedure began with the positioning of an N-SF6 glass cylinder, having a length of $l = (3,044 \pm 0,005)\,\text{cm}$, inside a magnetic field generated by a solenoid coil supplied by a current of $I = 1,5\,\text{A}$. A Hall probe was utilized to map the magnetic field distribution at spatial intervals of $\Delta x = 0,2\,\text{cm}$ fig [4].

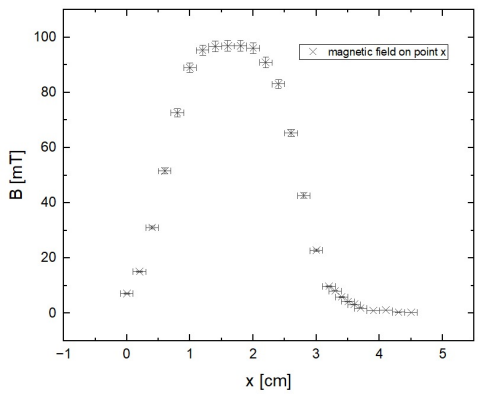

Figure 4: Measured magnetic flux with a Hall probe for calibration.

To derive the effective magnetic field, the raw data points were fitted using a sigmoidal regression function:

$$B(x) = A_2 - \frac{A_1 - A_2}{1 + \exp\left(\frac{x - x_0}{d}\right)} \tag{9}$$

Leveraging the field symmetry illustrated in fig 5, the effective magnetic flux density was computed via integration:

$$B_{eff} = \frac{1}{l}\int_0^l B(x)\,\mathrm{d}x \tag{10}$$

This calculation yields an effective field value of $B_{eff} = (68,9 \pm 1,8)\,\text{mT}$. To extend this calibration to other current levels, a proportionality factor α was established, defined as the ratio of the effective field to the peak plateau field:

$$\alpha = \frac{B_{eff}}{B_{max}} = 0,713 \pm 0,036 \tag{11}$$

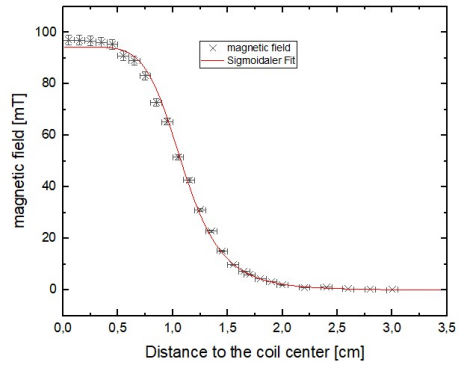

Figure 5: Folded data of magnetic flux for a regression analysis.

7.1.2 Measurement of Faraday Rotation

Additionally, the correlation between the applied current and the maximum magnetic flux density was characterized with the probe situated at the center of the uniform field region. To investigate the effective magnetic flux density along the cylinder for varying currents, the peak field values were recorded for each current setting. Uncertainty in the current values was estimated based on the specifications of the multimeter [Source 4], considering a range of 10 A.

As anticipated, the magnetic field exhibits a linear response to the current fig [6], consistent with the predictions of the Biot-Savart law. A linear regression of the form $B(I) = B_0 + \gamma I$ yields the following parameters:

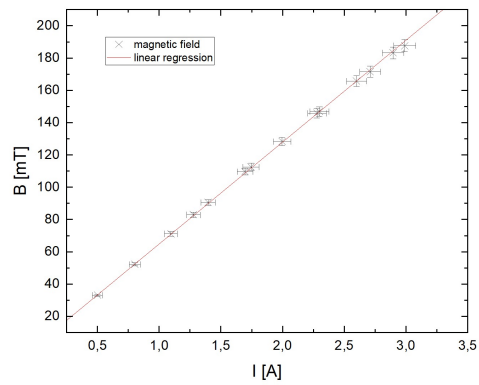

Figure 6: The data of current and magnetic flux density at the same center position.

$$B_0 = (1,70 \pm 0,28)\,\text{mT} \quad \text{and} \quad \gamma = (63,11 \pm 0,23)\,\frac{\text{mT}}{\text{A}}$$

Polarization rotation angles θ, were recorded for four distinct wavelength ranges, $\lambda_1 = (400 - 560)\,\text{nm}$, $\lambda_2 = (560 - 630)\,\text{nm}$, $\lambda_3 = (360 - 460)\,\text{nm}$ and $\lambda_4 = (480 - 570)\,\text{nm}$, across a current sweep ranging from

0 to 3 A. The effective magnetic field B_{eff} for each step was derived using the previously determined γ and α factors. The light source utilized continuous illumination filtered through specific color filters.
The resulting Faraday rotation data, visualized in fig [7], was fitted with linear functions to determine the angular rotation per unit field:

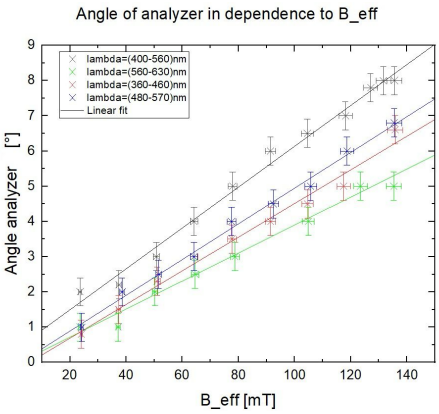

Figure 7: Beam intensity minimum for different B_{eff}.

Due to the Faraday effect, the plane of polarization is rotated as light propagates through the glass cylinder under the influence of an effective magnetic flux density. To quantify this rotation, the analyzer was adjusted to achieve minimum beam intensity. The results verify a linear relationship regarding the effective magnetic flux density and demonstrate a clear dependence on the wavelength used.

$$\frac{\theta}{B_{\lambda 1}} = (0,0580 \pm 0,0019)\,\frac{^\circ}{\mathrm{mT}} \tag{12}$$

$$\frac{\theta}{B_{\lambda 2}} = (0,0398 \pm 0,0025)\,\frac{^\circ}{\mathrm{mT}} \tag{13}$$

$$\frac{\theta}{B_{\lambda 3}} = (0,0478 \pm 0,0022)\,\frac{^\circ}{\mathrm{mT}} \tag{14}$$

$$\frac{\theta}{B_{\lambda 4}} = (0,0507 \pm 0,0015)\,\frac{^\circ}{\mathrm{mT}} \tag{15}$$

By applying Eq. [3] to the experimentally derived slopes, the Verdet constants were calculated as follows:

$$V_{\lambda 1} = (19,05 \pm 0,62)\,\frac{^\circ}{\mathrm{Tm}} \tag{15}$$

$$V_{\lambda 2} = (13,06 \pm 0,82)\,\frac{^\circ}{\mathrm{Tm}} \tag{16}$$

$$V_{\lambda 3} = (15,69 \pm 0,73)\,\frac{^\circ}{\mathrm{Tm}} \tag{17}$$

$$V_{\lambda 4} = (16,66 \pm 0,48)\,\frac{^\circ}{\mathrm{Tm}} \tag{18}$$

As expected, these constants depend on the wavelength and decrease with increasing wavelength with one exception. Uncertainties were computed via Gaussian error propagation. The smallest Verdet constant is not determined for the smallest wavelength filter, this deviation is likely attributable to thermal effects, as the magnet heats up, the magnetic susceptibility decreases, leading to a weaker actual field than the calibration assumes, thereby causing an underestimation of the Faraday rotation.

8 Conclusion

In this experiment, light from an incandescent lamp was linearly polarized using a polarizing filter and its intensity was measured after passing through y rotatable analyzer. This measurement onfirmed that the intensity of the linearly polarized light is proportional to the cosine of the angle between the polarizing filter and the analyzer. The intensity behaves as predicted by Malus's law.
The unpolarized light from the incandescent lamp was then passed through an anisotropic calcite crystal. Two beams emerging from the crystal were identified on a screen. The screen was shielded from the direct light of the incandescent lamp. A polarizing filter was used as an analyzer to determine the polarization of the two light beams. The result was that the ordinary and extraordinary beams are linearly polarized, but shifted relative to each other by $\Delta\theta = (90,0 \pm 1,4)^\circ$.
The operating principle of quarter-wave plates was investigated. If the angle between the polarization of the light and the axis of the plate is $\alpha = [0, \frac{\pi}{2}]^\circ$, the light remains linearly polarized. No phase shift occurs. If the angle is $\alpha = \frac{\pi}{4}$, both axes are excited equally, and a phase shift of $\Delta\phi = \frac{\pi}{2}$ occurs. The light is circularly polarized. If the angle between $\alpha = 0$ and $\alpha = \frac{\pi}{2}$ and the plate not $\alpha = \frac{\pi}{4}$, the optical axes are not excited equally. Elliptically polarized light is produced.
To determine the rotation angle of sugar, a penumbral polarimeter was used. The penumbral polarimeter used is $\Psi = (10,36 \pm 0,61)^\circ$. A solution with a sugar concentration of $\rho_0 = (10,3 \pm 2,1) * 10^{-2}\frac{kg}{L}$ was analyzed, and the penumbral angle was plotted against the wavelength. No dispersion relationship could be established. In the next step, the unknown sugar concentration of a solution was determined. It turned out that the determined sugar concentration was consistent with the literature value.

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